

ASSIGNMENT 5

Multi-Modal University Chatbot System

(CLO-3)

Human Computer Interaction

SE305T & MD445T (Spring-26)
BSSE23 & BSCE22

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Executive Summary

This report presents the successful completion of a comprehensive Multi-Modal University Chatbot System designed for Punjab University. The system integrates four critical modules: Spam Detection, Intent Classification, Response Generation, and Voice Interaction capabilities. The chatbot serves as an intelligent front-end interface that automates handling of repetitive university-related queries regarding admissions, fees, courses, and academic schedules.

The implementation successfully fulfills all assignment requirements and provides a practical solution to reduce administrative workload. The system achieves accurate spam detection using both rule-based dictionary matching and machine learning techniques. Intent classification leverages keyword mapping combined with TfidfVectorizer and Naive Bayes models to identify user query categories with high confidence. The response generation module provides both static predefined responses and dynamic context-aware answers based on specific keywords within queries.

A standout feature is the robust voice interaction capability, supporting both Speech-to-Text (STT) and Text-to-Speech (TTS) functionality. This enables users with accessibility needs and provides a more natural, conversational interface. The system operates in three modes: Text Input Mode, Voice Input Mode, and Hybrid Mode, allowing users to seamlessly switch between interaction methods. The chatbot presents comprehensive confidence analysis for spam detection, intent recognition, and unknown query classification, providing transparency in decision-making.

The team successfully distributed responsibilities based on individual strengths, with each member completing their assigned modules with high quality. Faiqa Arshad developed the

robust spam detection and intent classification systems using both traditional and machine learning approaches. Muhammad Saqlain Tahir implemented sophisticated response generation logic and integrated all modules into a cohesive system with proper menu-driven interface. Zunaira Abdul Aziz engineered the voice interaction capabilities, enabling Speech-to-Text and Text-to-Speech features with real-time integration throughout the application.

This chatbot demonstrates practical application of Human-Computer Interaction principles, combining natural language processing, machine learning, and voice technology to create an accessible, user-friendly system that addresses real university administrative challenges. The modular design ensures maintainability and extensibility for future enhancements.

Work Distribution Summary

The following table outlines the percentage contribution of each team member across different modules and components:

Module/Task	Faiqa Arshad	Saqlain Tahir	Zunaira Aziz
Module 1: Spam Detection	100%	0%	0%
Module 2: Intent Classification	100%	0%	0%
Module 3: Response Generation	0%	100%	0%
Module 4: Voice Interaction	0%	0%	100%
Integration & Testing	0%	100%	0%
Total Contribution	33.3%	33.3%	33.3%

Individual Contributions

Faiqa Arshad (BSSE23028) – Modules 1 & 2 Implementation

Faiqa Arshad successfully implemented Module 1 (Spam Detection) and Module 2 (Intent Classification), both critical components of the chatbot system. Her contributions demonstrate strong expertise in machine learning and natural language processing techniques. In Module 1, she created a comprehensive SpamDetector class that implements both rule-based dictionary matching and sophisticated machine learning approaches. The spam detection system includes an extensive categorized dictionary with over 45 spam keywords organized into categories like Exaggerated Claims, Urgency Pressure, Shady Behavior, and Jargon Spam. This multi-layered approach provides both immediate filtering capability and probabilistic analysis using TfidfVectorizer and Multinomial Naive Bayes classifier. The system achieves high accuracy in identifying irrelevant or malicious queries while avoiding false positives on legitimate university-related questions. For Module 2, Faiqa developed an intelligent IntentClassifier that recognizes four primary university intents: Admission, Fee, Courses, and Schedule. She expanded the keyword dictionary to include variations of terminology that students commonly use when asking about these topics. The implementation combines keyword matching with

ML-based probability estimation, allowing the system to handle variations in user phrasing. Her work ensures that student queries are correctly categorized for appropriate response generation. Faiqa's contributions significantly improve the system's ability to understand user intent and provide relevant assistance while protecting against spam and irrelevant content.

Muhammad Saqlain Tahir (BSSE23050) – Module 3 & System Integration

Muhammad Saqlain Tahir completed Module 3 (Response Generation) and served as the integration lead, ensuring all components work seamlessly together as a cohesive system. His role was crucial in transforming individual modules into a functional, user-friendly chatbot application. In Module 3, he designed the ResponseGenerator class that provides meaningful, context-aware responses to student queries. The implementation includes static responses for each intent category with relevant information about PUCIT, IBIT, and Punjab University programs, fees, and schedules. Beyond static responses, Saqlain implemented sophisticated dynamic response logic that adapts answers based on specific keywords within queries. For example, the system differentiates between morning and evening program fees, provides IBIT-specific information when detected, and offers program-specific course details. This dynamic approach ensures responses are tailored to student needs rather than generic. As the integration lead, Saqlain architected the main PU_IT_Chatbot controller class that orchestrates all modules into a unified pipeline. He implemented the complete menu-driven interface allowing users to select between Text Input Mode, Voice Input Mode, and Hybrid Mode. His integration work ensures proper data flow: spam detection first, followed by intent classification, and finally response generation. He also implemented sophisticated confidence scoring mechanisms that display transparency in decision-making to users. Saqlain's contributions transform separated modules into a production-quality system that is both user-friendly and technically robust.

Zunaira Abdul Aziz (BSSE23058) – Module 4 Voice Interaction

Zunaira Abdul Aziz engineered the complete Module 4 (Voice Interaction), which represents two distinct feature sets: Speech-to-Text input and Text-to-Speech output. This module significantly enhances the chatbot's accessibility and user experience. The voice interaction implementation includes Task 1 (Speech-to-Text), where users can speak queries using their microphone, and the system converts audio to text using Google's Speech Recognition API. Zunaira implemented real-time audio capture with ambient noise adjustment and comprehensive error handling for scenarios like unclear audio, service unavailability, and timeout conditions. The system displays recognized text to users before processing, ensuring transparency and allowing users to verify correct transcription. For Task 2 (Text-to-Speech), she integrated pyttsx3 library to convert chatbot responses into natural-sounding speech output. The speech rate is configured appropriately at 165 words per minute for clear, comfortable listening. Task 3 integrates voice seamlessly throughout the entire pipeline: users can input via voice, the chatbot processes the query through spam detection and intent classification, and returns responses via speech output. Zunaira implemented the complete voice flow, enabling users with different interaction preferences and accessibility needs to use the system effectively. Her work on voice interaction demonstrates understanding of accessibility principles in HCI. The Hybrid Mode she enabled allows users to dynamically switch between text and voice input during the same session, providing maximum flexibility. Zunaira's contributions make the chatbot accessible to users with

visual impairments or those who prefer voice-based interaction, significantly broadening the system's usability.

Technical Implementation Details

Module 1: Spam Detection

The spam detection module employs a two-tier approach. First, rule-based dictionary matching checks queries against categorized spam keywords. Second, a machine learning pipeline combines TfidfVectorizer with Multinomial Naive Bayes to calculate probability scores. The system returns spam confidence metrics, spam category information, and handles edge cases gracefully.

Module 2: Intent Classification

Intent classification combines keyword matching for immediate identification with machine learning for uncertain cases. The system recognizes four primary intents (Admission, Fee, Courses, Schedule) and returns both intent label and confidence scores. Keyword expansions handle common variations in student terminology.

Module 3: Response Generation

The response generator provides both static predefined responses and dynamic context-aware answers. Static responses provide baseline information for each intent. Dynamic logic analyzes specific keywords to customize responses (morning/evening fees, IBIT-specific information, program details, etc.).

Module 4: Voice Interaction

Voice interaction utilizes SpeechRecognition library for STT with ambient noise adjustment and timeout handling. Text-to-speech uses pyttsx3 with configurable speech rate. The implementation supports real-time voice integration throughout the chatbot pipeline.

System Architecture

The main PU_IT_Chatbot controller orchestrates all modules, implementing a processing pipeline that first checks for spam, then classifies intent, and finally generates appropriate responses. The system calculates unknown confidence as a residual metric, providing transparency in all decisions. Three input modes support diverse user preferences and accessibility needs.

System Features & Capabilities

Input Modes

Text Input Mode: Users type queries directly via console. Voice Input Mode: Users speak queries, converted to text via speech recognition. Hybrid Mode: Users dynamically switch between text and voice within the same session.

Confidence Analysis

The system displays three confidence scores: Spam Confidence (probability of spam), Intent Confidence (accuracy of intent classification), and Unknown Confidence (residual uncertainty). This transparency helps users understand system decisions.

University-Specific Knowledge

The system provides accurate information about Punjab University programs (PUCIT, IBIT), fee structures for morning and evening programs, admission requirements, and academic schedules. Responses are customized based on program and delivery method.

Accessibility

Full voice interaction support makes the system accessible to users with visual impairments. Real-time text display of speech recognition output ensures deaf and hard of hearing users can verify understood content.

Testing & Results

Spam Detection Testing

Tested against 5+ predefined spam cases per requirements. System correctly identified promotional content, phishing attempts, and irrelevant queries. Achieved high accuracy without false positives on legitimate university questions. Dictionary-based detection provides immediate response, while ML component adds sophisticated probability estimation.

Intent Classification Testing

Successfully classified 10+ test queries with accurate intent detection. System handles variations like 'fee' vs 'cost', 'admission process' vs 'apply', 'course offerings' vs 'what programs'. Confidence scores accurately reflect classification certainty. Both keyword matching and ML approaches work in tandem.

Response Generation Testing

Static responses provide baseline information for all four intents. Dynamic response logic customizes answers for specific scenarios (e.g., morning vs evening fees, IBIT program details). System gracefully handles unknown queries with helpful guidance.

Voice Interaction Testing

Speech-to-text successfully converts voice input with appropriate error handling for unclear audio. Text-to-speech output is clear and at comfortable listening speed. Complete voice pipeline from voice input through processing to voice response works reliably in real-time.

Input Mode Testing

All three input modes (Text, Voice, Hybrid) function correctly. Menu system is intuitive and responsive. ESC key exit works on Windows systems. Continuous interaction mode supports multiple queries per session. System maintains state appropriately across mode switches.

Conclusion

This assignment successfully demonstrates the complete design and implementation of a sophisticated multi-modal chatbot system suitable for real-world university deployment. All four

core modules have been implemented with quality and integration. The system fulfills every requirement specified in the assignment brief.

The team's collaborative approach, with clear responsibility division based on individual strengths, ensured efficient development and high-quality output. Faiqa's strong machine learning background enabled robust spam detection and intent classification. Saqlain's systems thinking and integration skills created a cohesive, user-friendly application. Zunaira's HCI focus resulted in accessible voice interaction features that significantly enhance usability.

Key achievements include accurate multi-layer spam detection, reliable intent classification handling query variations, sophisticated dynamic response generation, and comprehensive voice interaction capabilities. The confidence analysis system provides transparency in decision-making, and the three-mode input system accommodates diverse user preferences and accessibility requirements.

Future enhancements could include database integration for tracking queries, natural language generation for more varied responses, integration with official university systems for real-time information, and multilingual support for international students. The modular architecture ensures the system can be easily extended with these improvements. This chatbot prototype demonstrates practical application of HCI principles and NLP techniques to solve genuine university administrative challenges.